APPLICANT:	FFVP	CRI:	
PROJECT NO.:		ISSUE:	01
PROJ. TITLE:		DATE:	05/06/2026
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APPLICABILITY:		Next ACTION:	xxxxx

SUBJECT:	Measured-Usage Fatigue Monitoring for Composite Sailplanes
CATEGORY:	Means of Compliance (MoC)
REQUIREMENTS:	CS 22.627 CS 22.1529 CS 22.1585
ADVISORY MATERIAL:	EASA CM-S-006 Issue 01
PRIMARY PANEL:	03
SECONDARY PANEL:	05
PUBLIC CONSULTATION	Not applicable

IDENTIFICATION OF ISSUE

CS-22 currently recognizes several Acceptable Means of Compliance to demonstrate fatigue strength under CS 22.627, including full-scale fatigue testing based on an accepted reference load spectrum, or static overload testing at ultimate load plus additional safety margin.
In all cases, EASA grants a safe-life limit based on conservative certification assumptions intended to cover the approved type operation. The 12,000-hour limit applied to many composite sailplanes reflects both historical uncertainty regarding composite fatigue behaviour and the conservatism of the reference utilization spectrum, which represents a full operational usage envelope and may exceed the measured operational usage of a significant portion of the fleet. Operational experience, published composite data, full-scale fatigue tests, and analytical comparisons indicate that this limit is conservative with respect to typical measured operational usage.
Similar measured-usage principles have been used in aviation, including Structural Health Monitoring and Individual Aircraft Tracking, particularly in military aviation where aircraft may experience missions significantly different from the original design assumptions.
The proposed measured-usage fatigue monitoring method does not claim direct transferability from SHM or IAT practice. The relevant precedent is the general principle that measured operational usage can be compared with a conservative reference usage assumption, provided the method is controlled, conservative, and supported by inspections.

Measured-usage fatigue monitoring offers an alternative solution by recording representative in-service usage parameters, processing measured operational data, and calculating a fatigue severity index that can be compared with a reference index built upon an agreed reference spectrum. Such a usage-severity index should not be considered as a direct measurement of physical composite damage but can be used as one element of an approved process to support operation beyond the originally established safe-life limit, provided the applicable continued-airworthiness and inspection requirements are satisfied.

This CRI establishes the Means of Compliance for measured-usage fatigue monitoring as an alternative fatigue-life management method under CS 22.627, consistent with EASA CM-S-006 and without requiring any change to the certified structural loads, certified flight envelope, original structural substantiation, or Type Certificate structural documentation. The proposed approach does not establish a new type-level structural fatigue life limit. On an individual-aircraft basis, it provides an alternative means to support operation beyond the Type Certificate safe-life limit while ensuring that the approved reference fatigue exposure limit is not exceeded.

DISCUSSION

EASA POSITION (Issue 1, dated xx-xxx-yyyy) (drafted by applicant)
EASA may accept measured-usage fatigue monitoring as an Alternative Means of Compliance to CS 22.627 for individual composite sailplanes certificated under CS-22 with an established safe-life limit, provided all requirements of Appendix A are met.
This acceptance: (a) applies on a per-aircraft serial-number basis; (b) does not allow any modification to the fatigue metric value associated with the certified safe-life basis, the certified flight envelope limits, the certified structural load limits, or the original structural substantiation; and (c) is valid only if the Instructions for Continued Airworthiness associated with the monitoring system, including required inspections, are satisfied.

APPLICANT'S POSITION (Issue xx, dated xx-MAY-2026)

The FFVP agrees with the EASA position and will use the MoC for the compliance demonstration.

CONCLUSION (Issue xx, date)

TBD

CRI REVISION RECORD

Issue	Date	CRI Changes
1		Initial issue

Appendix A – Proposed Means of Compliance for Measured-Usage Fatigue Monitoring

For a glider with an established safe-life limit, a system for monitoring in-service usage may support operation beyond the originally established safe-life limit, provided that the metric used to monitor fatigue remains below the approved reference metric value associated with the certified safe-life limit and that the applicable continued-airworthiness provisions are satisfied.

The method does not modify the certified flight envelope, the certified structural load limits, or the original structural substantiation, and is not intended to measure the actual physical damage state of the glider structure.

The intent of this Appendix is to provide additional Means of Compliance / Interpretative Material for CS 22.627, enabling compliance solutions based on measured in-service usage of glider structures.

• Fatigue-severity metric:
A metric should be established to represent glider composite-structure fatigue severity and to allow comparison between two glider usage spectra. An Equivalent Fatigue Index calculated using Palmgren-Miner and Basquin models is considered acceptable. The Equivalent Fatigue Index is a comparative usage-severity metric and shall not be interpreted as a direct measurement of physical damage in the composite structure.

• Reference spectrum:
A reference spectrum should be selected to represent the envelope of possible glider usages within the approved applicability domain and should be recognized as a means of compliance. For Utility-category operation, the Kossira & Reinke Utility spectrum is considered acceptable, consistent with EASA CM-S-006.

• Reference metric value:
A reference metric value should be established from the selected reference spectrum and should correspond to the exposure limit that an individual glider tracked by measured-usage fatigue monitoring must not exceed. If the Palmgren-Miner model is used, it is acceptable to define the metric such that an exposure equal to three times the certified safe-life limit corresponds to a metric value of one (1). In that case, the metric value corresponding to the certified safe-life limit is one third of the metric value associated with the safety-factored exposure. This normalization shall not be interpreted as evidence that the structure would fail at that exposure; it is a conservative reference normalization for the measured-usage comparison.

• Real usage spectrum:
A real usage spectrum should be established to represent the load variations and fatigue-relevant usage actually sustained by an individual glider. To allow comparison between the reference spectrum and the real usage spectrum, the real usage spectrum should be developed using a methodology consistent with the one used to establish the reference spectrum. When a Kossira & Reinke reference spectrum is used, the Kossira & Reinke counting methodology is considered acceptable. Any period for which valid measured usage data are not available shall be accounted for using an approved conservative substitution method, such as reference-spectrum usage for the corresponding flight time.

• Measurement system:
To establish a real usage spectrum, an acquisition system capable of recording parameters representative of the stress variations sustained by an individual glider should be used. A recording system based on accelerations is considered acceptable, provided a justified relationship can be established between the measured accelerations and the fatigue-relevant stress variations, and provided the limits of this relationship are identified.

• Composite fatigue parameter:
When the metric is developed using Palmgren-Miner and Basquin models, the Basquin parameter

should be based on recognized composite-structure fatigue data representative of composite primary structures used in gliders. The parameter should be selected conservatively, in order to cover the range of composite structural configurations found in main glider structures. Recognized composite fatigue databases and reports, including Sandia National Laboratories / MSU / DOE wind-blade composite fatigue data, may be used as acceptable sources of Basquin parameters, provided the selected value is justified as conservative for the intended glider structural application.

• Reference-versus-real comparison:
When the reference metric, real usage metric, and spectrum-comparison model are established, they may be used to assess eligibility for operation beyond the originally established safe-life limit for an individual glider. The method should ensure that the cumulative real usage metric, including any conservative substitution required by the approved process, does not exceed the approved reference metric value. Using an Equivalent Fatigue Index is considered acceptable to compare real usage against reference usage and to ensure that the approved limit is not exceeded. The method is based on comparison of global usage severity against an accepted reference usage condition and is not intended to replace inspections or assessments addressing local defects, repairs, bonds, fittings, ageing effects, or other airworthiness-significant conditions.

• Instructions for Continued Airworthiness and operational control:
Instructions for Continued Airworthiness should be developed so that the benefits and limitations of measured-usage fatigue monitoring are correctly taken into account in service. These instructions should include AFM, AMM, AMP, or equivalent document updates as appropriate, to allow the monitoring benefits to be used while ensuring that the monitoring limitations are respected before any Airworthiness Review or release-to-service decision.